

# FLORENT TARIOLLE

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*Engineering student seeking a 6-month final-year research internship in deep learning, starting in 2027.*

## EDUCATION

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| <b>INSA Rouen Normandy</b><br>Engineering degree, Computer Science and AI, expected 2027<br><i>Approx. U.S.-equivalent GPA: 3.7/4.0 overall; 4.0/4.0 latest semester.</i> | Rouen, France<br>2023 - 2027   |
| <b>University of Rouen Normandy</b><br>M.Sc. in Data Science, research-oriented MLAI track, expected 2027                                                                 | Rouen, France<br>2026 - 2027   |
| <b>Lycée Dupuy de Lôme</b><br>CPGE MP (Preparatory Classes for Grandes Écoles, Mathematics and Physics)                                                                   | Lorient, France<br>2021 - 2023 |

## PUBLICATIONS

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| <b>Opportunistic Target Selection</b><br>F. Tariolle and F. Yger. <i>CAp 2026 (Conférence sur l'Apprentissage automatique)</i> .                                                                                                                                                                                                                                                                                                                       | May 2026 |
| <ul style="list-style-type: none"><li>Introduced a lightweight wrapper for score-based black-box adversarial attacks using early class confidence signals to reduce class drift in undirected random-search attacks.</li><li>Improved attacks by up to 27 percentage points in success rate and a 43% relative reduction in censored-mean iterations on ResNet-50, validated across 4,500 ImageNet attack runs on five standard classifiers.</li></ul> |          |

## EXPERIENCE

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| <b>InterDigital</b><br><i>Deep Learning Research Intern</i>                                                                                                                                                                                                       | Rennes, France<br>May 2026 - Present |
| <ul style="list-style-type: none"><li>Working on deep learning methods for next-generation video compression, focusing on VVC (H.266) in-loop filtering and decoder-side filter selection to improve reconstructed video quality and coding efficiency.</li></ul> |                                      |
| <b>Enedis</b><br><i>Machine Learning Intern</i>                                                                                                                                                                                                                   | Rouen, France<br>Jan 2026 - May 2026 |
| <ul style="list-style-type: none"><li>Working on deep generative models and large-scale pipelines to simulate high-frequency time series for 38 million customers, supporting a digital twin of the French electricity grid.</li></ul>                            |                                      |

## RESEARCH PROJECTS

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| <b>DashVMC: Real-Time Discrete World Model Control in Geometry Dash</b><br><i>Independent research project; manuscript in preparation.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Feb 2026 - Present |
| <ul style="list-style-type: none"><li>Built a real-time Geometry Dash Vision-Model-Controller system with FSQ tokenization, action-conditioned transformer dynamics, and an actor-critic trained from behavioural cloning plus PPO in latent FSQ-token rollouts.</li><li>Deployed the learned controller on live gameplay at 30 FPS; decoded pixel rollouts are used only for qualitative demos, including plausible level continuations from real gameplay prefixes.</li><li>Studied Structured Label Smoothing as an FSQ-neighbour prior; a matched IRIS/Pong diagnostic showed partial stability signals but no robust general improvement over cross-entropy.</li></ul> |                    |

## AWARDS

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| <b>Finalist, Hack the World(s) Hackathon</b><br><i>24-hour team project at a Yann LeCun-sponsored hackathon on JEPAs and world models</i>                                                                                              | Jun 2026 |
| <ul style="list-style-type: none"><li>Developed a geometry-aware joint-embedding EEG SSL system: strong SIGReg frozen-transfer baselines and Riemannian latent analysis showing structure in SPD covariance representations.</li></ul> |          |

## OPEN SOURCE

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| <b>Rose</b> (Active: 15K+ DAU)<br><i>Co-founder &amp; Lead Developer</i>                                                                                                                                                                                                                                                                                                                                                          | Sep 2025 - Present |
| <ul style="list-style-type: none"><li>Co-founded and led development of a high-traffic real-time customization tool used by 15K+ daily active users; coordinated a 6-person team and designed the Python backend, Cloudflare Workers WebSocket relay, and browser-side JavaScript integration.</li><li>Replaced OCR-based state extraction with deterministic DOM parsing, reducing latency and operational complexity.</li></ul> |                    |

## TECHNICAL SKILLS

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| <b>Programming:</b> Python, C++<br><b>AI &amp; ML:</b> PyTorch, CUDA; Deep Learning, Self-Supervised Learning, World Models, Model-Based RL and Planning<br><b>Languages:</b> French (native), English (TOEIC 970/990) |
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## REFERENCES

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|-------------------------------------------------|------------------------------------------------------------------|
| <b>Florian Yger</b><br><b>Frédéric Lefebvre</b> | florian.yger@insa-rouen.fr<br>frederic.lefebvre@interdigital.com |
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